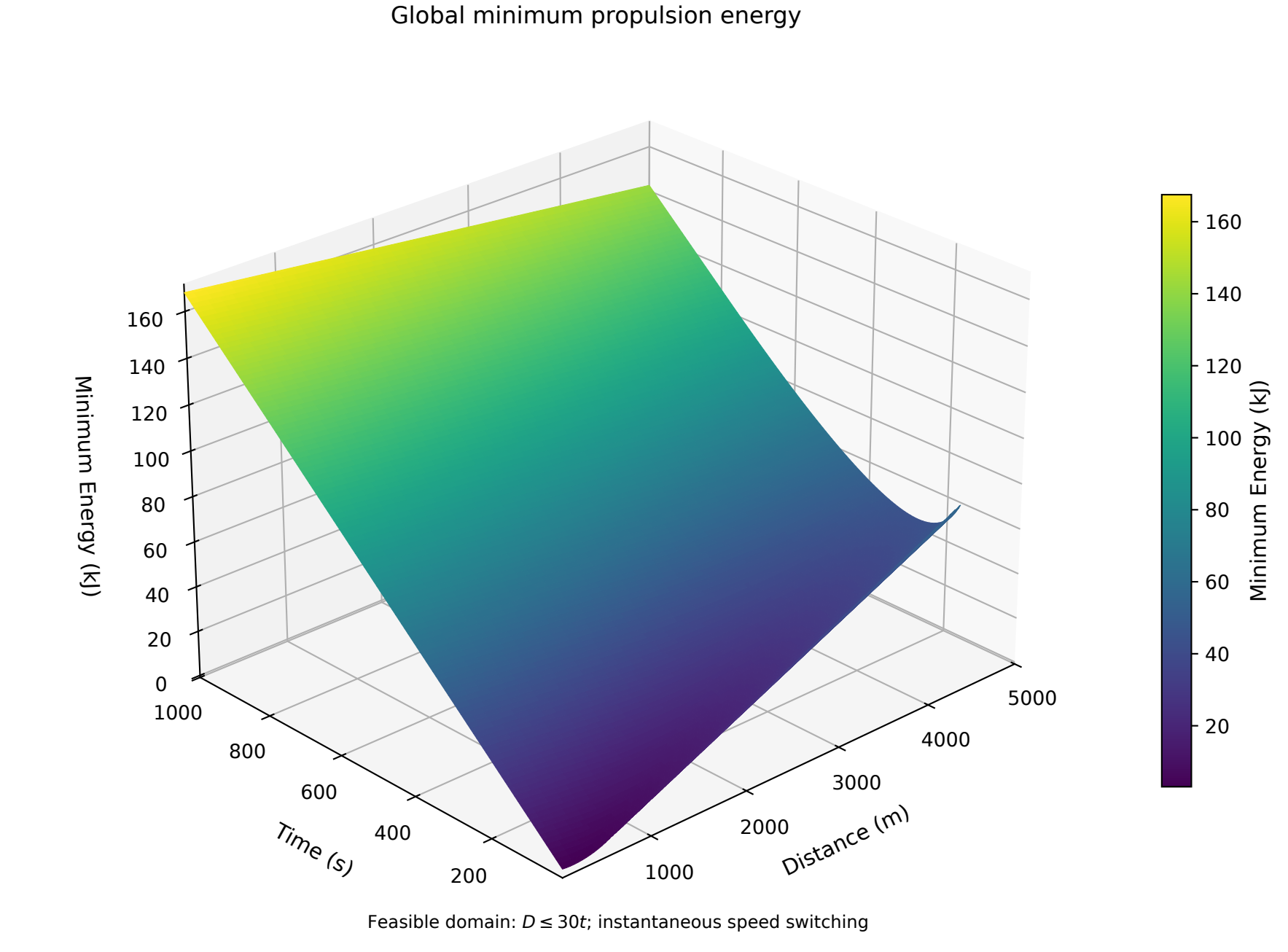

Global minimum propulsion energy
Minimum Energy (kJ)
0
20
40
60
80
100
120
140
160
Time (s)
1000
800
600
400
200
Distance (m)
1000
2000
3000
4000
5000
Minimum Energy (kJ)
20
40
60
80
100
120
140
160
Feasible domain: $D \leq 30t$; instantaneous speed switching